Expanded comparison: complete choice-model equations

Thirteen active models | Shared definitions + model-specific recursions | One-page reference, A0 landscape

Notation, observations and parameter domains

Session d contains task trials $t = 1, \dots, T_d$; the session index is suppressed below. $s_t > 0$: physical duration (seconds); X_t: internal measurement (seconds). $A_t = y_t \in \{0, 1\}$: observed choice; $C_t \in \{0, 1\}$: true stimulus category. 0 = short, 1 = long. $F_t \in \{0, 1\}$: valid category feedback. Define

$$U_t = \mathbf{1}\{A_t \text{ finite and binary}\}, \quad K_t = \mathbf{1}\{C_t \text{ finite and binary}\}, \quad D_t = \mathbf{1}\{s_t \text{ finite, } s_t > 0\}.$$

$F_t = 1$ requires $K_t = 1$. Undefined choices are omitted from the choice likelihood, but history/latent updates follow their own gates. All state variables below are *pre-choice*; a superscript + means post-feedback.

$$\operatorname{sigm}(x) = \frac{1}{1 + e^{-x}}, \quad \operatorname{logit}(p) = \log \frac{p}{1 - p}, \quad \operatorname{softplus}(x) = \log(1 + e^{x}).$$

With unconstrained raw coordinates denoted by tildes,

$$\begin{aligned} w &= \operatorname{softplus}(\bar{w}) + 10^{-10} > 0, & s_0 &= \operatorname{softplus}(\bar{s}_0) + 10^{-10} > 0, \\ \ell_k &= \operatorname{sigm}(\bar{\ell}_k) \in (0, 1), & \kappa &= e^{\overline{\log \kappa}} > 0, \\ \alpha &= \operatorname{sigm}(a) \in (0, 1), & \eta &= \operatorname{sigm}(e) \in (0, 1), \\ \delta_0 &= \operatorname{softplus}(\bar{\delta}_0) + 10^{-10} > 0, & b_h &= \operatorname{softplus}(\bar{b}_h) + 10^{-10} > 0. \end{aligned}$$

$k \in \{0, 1\}$; $\ell_0 = \ell_{\text{short}}$, $\ell_1 = \ell_{\text{long}}$. $a, a_h, e \in \mathbb{R}$; e is the registered `eta_logit`. w is a variance coefficient, s_0 a standard deviation; κ is the decision-cost ratio. a_h is the biased hazard logit (at run length 23 for dynamic models). Parameters are shared across sessions within each fitted window. Each model has its own fitted values.

Shared sensory readout $\mathcal{B}(s_t, \rho_t)$

Empirical task moments use all presented real stimuli of category k, including omitted-choice trials; choices/outcomes do not determine these fixed moments:

$$\mu_k = \frac{1}{N_k} \sum_{i:C_i = k} s_i, \qquad \tau_k^2 = \frac{1}{N_k} \sum_{i:C_i = k} (s_i - \mu_k)^2.$$

These are population variances (ddof = 0), evaluated using the task-moment source registered for the subject.

$$\begin{aligned} X_t \mid s_t &\sim \mathcal{N}(s_t, \sigma_t^2), & \sigma_t &= \sqrt{w s_t^2 + s_0^2} > 0, \\ f_k(x) &= \mathcal{N}(x; \mu_k, v_k), & v_k &= \tau_k^2 + w\mu_k^2 + s_0^2, \\ \bar{\rho}_t &= \operatorname{clip}(\rho_t, 10^{-10}, 1 - 10^{-10}), \\ g_t(x) &= \log \frac{f_1(x)}{f_0(x)} + \operatorname{logit}(\bar{\rho}_t) - \log \kappa \\ &= a_2 x^2 + a_1 x + a_{0t}, \\ a_2 &= \frac{1}{2v_0} - \frac{1}{2v_1}, \qquad a_1 = \frac{\mu_1}{v_1} - \frac{\mu_0}{v_0}, \\ a_{0t} &= \frac{\mu_0^2}{2v_0} - \frac{\mu_1^2}{2v_1} + \frac{1}{2} \log \frac{v_0}{v_1} + \operatorname{logit}(\bar{\rho}_t) - \log \kappa, \\ \mathcal{R}_t &= \{x \in \mathbb{R} : g_t(x) > 0\}, \\ \mathcal{B}(s_t, \rho_t) &= \Pr(X_t \in \mathcal{R}_t \mid s_t) = \int_{\mathcal{R}_t} \frac{e^{-(x - s_t)^2/(2\sigma_t^2)}}{\sqrt{2\pi}\sigma_t} \, dx. \end{aligned}$$

The Gaussian f_k is the model's category approximation; the measurement distribution is integrated over *all real* x, including negative tails. A tie $g_t(x) = 0$ chooses short.

Every analytical decision-region case

Let $H_t(x) = \Phi((x - s_t)/\sigma_t)$, where Φ is the standard normal CDF. Let $\zeta = 10^{-12}$, $\Delta_t = a_1^2 - 4a_2 a_{0t}$. The implemented branch priority is:

$$\mathcal{B}(s_t, \rho_t) = \begin{cases} \mathbf{1}\{a_{0t} > 0\}, & |a_2| \le \zeta, \ |a_1| \le \zeta, \\ 1 - H_t(-a_{0t}/a_1), & |a_2| \le \zeta, \ a_1 > \zeta, \\ H_t(-a_{0t}/a_1), & |a_2| \le \zeta, \ a_1 < -\zeta, \\ \mathbf{1}\{a_2 > 0\}, & |a_2| > \zeta, \ \Delta_t \le \zeta, \\ H_t(L_t) + 1 - H_t(R_t), & |a_2| > \zeta, \ \Delta_t > \zeta, \ a_2 > 0, \\ H_t(R_t) - H_t(L_t), & |a_2| > \zeta, \ \Delta_t > \zeta, \ a_2 < 0. \end{cases}$$

For the last two cases,

$$x_{\pm,t} = \frac{-a_1 \pm \sqrt{\max(\Delta_t, 0)}}{2a_2}, \qquad L_t = \min(x_{-,t}, x_{+,t}), \quad R_t = \max(x_{-,t}, x_{+,t}).$$

$\Delta_t < -\zeta$: no real roots; $|\Delta_t| \le \zeta$: double/near-double root $-a_1/(2a_2)$. Both give the constant probability shown (a point has zero Gaussian mass). No positive-root-only restriction is imposed.

Shared category-dependent lapse and likelihood

For every model, m_t denotes its pre-lapse long-choice probability:

$$\epsilon_t = \ell_0 \mathbf{1}\{C_t = 0\} + \ell_1 \mathbf{1}\{C_t = 1\}, \qquad p_t = \Pr(y_t = 1) = (1 - \epsilon_t) m_t + \frac{\epsilon_t}{2}.$$

Thus current category selects the lapse rate even in the four history-only models. It does not select their history state or an inferred block.

$$\bar{p}_t = \operatorname{clip}(p_t, 10^{-12}, 1 - 10^{-12}), \qquad L_t^{\text{num}} = \bar{p}_t^{\,y_t} (1 - \bar{p}_t)^{1 - y_t}.$$

Here L_t^{num} is the numerically evaluated Bernoulli contribution. Clipping is numerical: saved p_t and m_t remain unclamped. The common objective is given in the last column.

Models 1–4: history without sensory readout

h_t is a scalar long-choice tendency, reset to $h_1 = 1/2$ in every session. Updates occur *after* scoring trial t. An inactive gate holds h_t unchanged; elapsed omitted trials cause no history decay.

1. Single-trial WSLS

2 parameters

`Pure_WSLS_v1` [`wslsSingle_lapse`]

$$\Theta = (\ell_0, \ell_1), \quad \alpha \equiv 1, \quad h_1 = \tfrac{1}{2}, \quad m_t = h_t,$$

$$p_t = (1 - \epsilon_t) h_t + \epsilon_t/2, \qquad h_{t+1} = \begin{cases} C_t, & F_t = 1, \\ h_t, & F_t = 0. \end{cases}$$

This is the last feedback-valid category, not necessarily category $t - 1$. In the binary task, the rewarded side means stay after a correct choice and switch after an incorrect choice.

2. Integrated WSLS history

3 parameters

`Extended_WSLS_v1` [`wslsHistory_lapse`]

$$\Theta = (\ell_0, \ell_1, a), \quad \alpha = \operatorname{sigm}(a), \quad h_1 = \tfrac{1}{2}, \quad m_t = h_t,$$

$$p_t = (1 - \epsilon_t) h_t + \epsilon_t/2, \qquad h_{t+1} = \begin{cases} h_t + \alpha(C_t - h_t), & F_t = 1, \\ h_t, & F_t = 0. \end{cases}$$

3. Single-trial action repetition

2 parameters

`Pure_action_v1` [`actionSingle_lapse`]

$$\Theta = (\ell_0, \ell_1), \quad \alpha \equiv 1, \quad h_1 = \tfrac{1}{2}, \quad m_t = h_t,$$

$$p_t = (1 - \epsilon_t) h_t + \epsilon_t/2, \qquad h_{t+1} = \begin{cases} A_t, & U_t = 1, \\ h_t, & U_t = 0. \end{cases}$$

The state equals the last valid observed action; feedback is irrelevant to this update.

4. Integrated action history

3 parameters

`Extended_action_v1` [`actionHistory_lapse`]

$$\Theta = (\ell_0, \ell_1, a), \quad \alpha = \operatorname{sigm}(a), \quad h_1 = \tfrac{1}{2}, \quad m_t = h_t,$$

$$p_t = (1 - \epsilon_t) h_t + \epsilon_t/2, \qquad h_{t+1} = \begin{cases} h_t + \alpha(A_t - h_t), & U_t = 1, \\ h_t, & U_t = 0. \end{cases}$$

Models 5–9: sensory evidence and simple history

Define the common sensory/lapse parameter vector $\Theta_S = (w, s_0, \ell_0, \ell_1, \kappa)$.

5. Sensory baseline (M0)

5 parameters

`M0_sensory_v1` [`sensory_lapse`]

$$\Theta = \Theta_S, \qquad \rho_t \equiv \tfrac{1}{2}, \qquad m_t = \mathcal{B}(s_t, \tfrac{1}{2}),$$

$$p_t = (1 - \epsilon_t) \mathcal{B}(s_t, \tfrac{1}{2}) + \epsilon_t/2.$$

There is no adaptive state or history update.

6. Sensory + integrated action prior

6 parameters

`M1_action_kernel_v1` [`sensory_actionHistory_lapse`]

$$\Theta = (\Theta_S, a), \quad \alpha = \operatorname{sigm}(a), \quad h_1 = \tfrac{1}{2}, \quad \rho_t = h_t,$$

$$m_t = \mathcal{B}(s_t, h_t), \qquad p_t = (1 - \epsilon_t) \mathcal{B}(s_t, h_t) + \epsilon_t/2,$$

$$h_{t+1} = \begin{cases} h_t + \alpha(A_t - h_t), & U_t = 1, \\ h_t, & U_t = 0. \end{cases}$$

7. Sensory + integrated category/WSLS prior

6 parameters

`M1_stimulus_kernel_v1` [`sensory_wslsHistory_lapse`]

$$\Theta = (\Theta_S, a), \quad \alpha = \operatorname{sigm}(a), \quad h_1 = \tfrac{1}{2}, \quad \rho_t = h_t,$$

$$m_t = \mathcal{B}(s_t, h_t), \qquad p_t = (1 - \epsilon_t) \mathcal{B}(s_t, h_t) + \epsilon_t/2,$$

$$h_{t+1} = \begin{cases} h_t + \alpha(C_t - h_t), & F_t = 1, \\ h_t, & F_t = 0. \end{cases}$$

Models 6–7 change the Bayesian *prior*; they do not mix history into its output.

8. Sensory + previous-action output bias

6 parameters

`M1_previous_action_bias_v1` [`sensory_actionSingle_lapse`]

$$\Theta = (\Theta_S, e), \quad \eta = \operatorname{sigm}(e), \quad \alpha \equiv 1, \quad h_1 = \tfrac{1}{2},$$

$$\rho_t = \tfrac{1}{2}, \quad m_t = (1 - \eta) \mathcal{B}(s_t, \tfrac{1}{2}) + \eta h_t,$$

$$p_t = (1 - \epsilon_t)\big[(1 - \eta) \mathcal{B}(s_t, \tfrac{1}{2}) + \eta h_t\big] + \epsilon_t/2, \quad h_{t+1} = \begin{cases} A_t, & U_t = 1, \\ h_t, & U_t = 0. \end{cases}$$

9. Sensory + previous-category/WSLS output bias

6 parameters

`M1_previous_wsls_bias_v1` [`sensory_wslsSingle_lapse`]

$$\Theta = (\Theta_S, e), \quad \eta = \operatorname{sigm}(e), \quad \alpha \equiv 1, \quad h_1 = \tfrac{1}{2},$$

$$\rho_t = \tfrac{1}{2}, \quad m_t = (1 - \eta) \mathcal{B}(s_t, \tfrac{1}{2}) + \eta h_t,$$

$$p_t = (1 - \epsilon_t)\big[(1 - \eta) \mathcal{B}(s_t, \tfrac{1}{2}) + \eta h_t\big] + \epsilon_t/2, \quad h_{t+1} = \begin{cases} C_t, & F_t = 1, \\ h_t, & F_t = 0. \end{cases}$$

Models 8–9 keep the sensory prior neutral; η mixes at the *probability output*, before lapse. An $\alpha = 1$ history fed into a Bayesian prior is a different construction.

Exact history kernel (models 2, 4, 6, 7)

For a driver V_t with gate G_t, let $j_1 < \cdots < j_{n_t} < t$ be its active update trials. $V = C, G = F$ for category history; $V = A, G = U$ for action history. The recurrence above is exactly

$$h_t = (1 - \alpha)^{n_t} \frac{1}{2} + \alpha \sum_{i=1}^{n_t} (1 - \alpha)^{n_t - i} V_{j_i}, \qquad n_t = \sum_{j < t} G_j.$$

Thus memory decays per valid update, not per elapsed task trial. For fixed $\alpha = 1$, use exact assignment to the last valid driver (or 1/2 if none).

Block-inference building blocks (models 10–13)

Latent states $z \in \{N, S, L\}$ mean neutral, short-biased and long-biased. All M2 prototypes are static fitted subjective category rates:

$$\theta_N = \tfrac{1}{2}, \quad \theta_S = \operatorname{sigm}(-\delta_0), \quad \theta_L = \operatorname{sigm}(\delta_0) = 1 - \theta_S.$$

For valid feedback define $E_t(z) = \theta_z^{C_t} (1 - \theta_z)^{1 - C_t}$. Feedback conditions the latent state; its emission likelihood is *not* added to the fitted choice objective.

10. Sensory + constant-hazard block inference

7 parameters

`M2_hmm_static_v1`

$$\Theta = (\Theta_S, a_h, \delta_0, b_h), \quad h_B = \operatorname{sigm}(a_h), \quad h_N = h_B/2, \quad \omega = 1/2.$$

Let $b_t(z) = \Pr(Z_t = z \mid \text{past feedback})$ be a column vector in (N, S, L) order:

$$b_1 = (1, 0, 0)^\top, \quad \rho_t = \sum_z b_t(z) \theta_z, \quad m_t = \mathcal{B}(s_t, \rho_t),$$

$$p_t = (1 - \epsilon_t) \mathcal{B}(s_t, \rho_t) + \epsilon_t/2,$$

$$b_t^+(z) = \begin{cases} \dfrac{b_t(z) E_t(z)}{\sum_{z'} b_t(z') E_t(z')}, & F_t = 1, \\ b_t(z), & F_t = 0, \end{cases} \qquad b_{t+1} = T^\top b_t^+,$$

$$T = \begin{pmatrix} 1 - h_N & (1 - \omega) h_N & \omega h_N \\ 0 & 1 - h_B & h_B \\ 0 & h_B & 1 - h_B \end{pmatrix}.$$

T has source rows and destination columns. Neutral is never re-entered; biased states alternate on switches.

11. Sensory + duration-dependent block inference

8 parameters

`M2_dynamic_block_length_v1`

$$\Theta = (\Theta_S, a_h, \delta_0, b_h, e), \quad h_B(r) = \operatorname{sigm}(a_h + b_h(r - 23)), \quad h_N(r) = h_B(r)/2.$$

Represented run length $r \in \{1, \dots, R\}$, $R = 65$; $r = R$ represents true run length $\ge R$ (a retained tail, not a forced switch). Let

$$\xi_t(z, r) = \Pr(Z_t = z, \min(\text{run length}_t, R) = r \mid \text{past feedback}),$$

$$\xi_1(z, r) = \mathbf{1}\{z = N, r = 1\}, \quad b_t(z) = \sum_{r=1}^{R} \xi_t(z, r), \quad \rho_t = \sum_z b_t(z) \theta_z,$$

$$m_t = \mathcal{B}(s_t, \rho_t), \qquad p_t = (1 - \epsilon_t) \mathcal{B}(s_t, \rho_t) + \epsilon_t/2.$$

The complete joint feedback and transition equations are

$$\xi_t^+(z, r) = \begin{cases} \dfrac{\xi_t(z, r) E_t(z)}{\sum_{z', u} \xi_t(z', u) E_t(z')}, & F_t = 1, \\ \xi_t(z, r), & F_t = 0. \end{cases}$$

Set $h_S(r) = h_L(r) = h_B(r)$, $\omega = 1/2$, and

$$S_t(z, r) = \xi_t^+(z, r)[1 - h_z(r)], \qquad W_t(z) = \sum_{r=1}^{R} \xi_t^+(z, r) h_z(r).$$

$$\begin{aligned} \xi_{t+1}(N, 1) &= 0, \\ \xi_{t+1}(S, 1) &= (1 - \omega) W_t(N) + W_t(L), \\ \xi_{t+1}(L, 1) &= \omega W_t(N) + W_t(S), \\ \xi_{t+1}(z, r) &= S_t(z, r - 1), \quad 2 \le r < R, \\ \xi_{t+1}(z, R) &= S_t(z, R - 1) + S_t(z, R). \end{aligned}$$

$\Sigma_{z,r} \xi_t(z, r) = 1$. Apply the transition on every task-trial step $t < T_d$, even without valid choice/feedback; no transition is needed after the final trial. Run length is inferred, never supplied from true block labels.

12. Dynamic block inference + previous-category/WSLS bias

9 parameters

`M2_dynamic_previous_wsls_bias_v1`

$$\Theta = (\Theta_S, a_h, \delta_0, b_h, e), \quad \eta = \operatorname{sigm}(e), \quad \alpha \equiv 1.$$

Use *all* model 11 prototype, hazard, initialization, prior, feedback and transition equations unchanged, and add the separate state

$$h_1 = \tfrac{1}{2}, \qquad h_{t+1} = \begin{cases} C_t, & F_t = 1, \\ h_t, & F_t = 0. \end{cases}$$

Replace only its pre-lapse readout and choice probability by

$$m_t = (1 - \eta) \mathcal{B}(s_t, \rho_t) + \eta h_t, \qquad p_t = (1 - \epsilon_t)\big[(1 - \eta) \mathcal{B}(s_t, \rho_t) + \eta h_t\big] + \epsilon_t/2.$$

13. Dynamic block inference + previous-action bias

9 parameters

`M2_dynamic_previous_action_bias_v1`

$$\Theta = (\Theta_S, a_h, \delta_0, b_h, e), \quad \eta = \operatorname{sigm}(e), \quad \alpha \equiv 1.$$

Use *all* model 11 prototype, hazard, initialization, prior, feedback and transition equations unchanged; its independent action state is

$$h_1 = \tfrac{1}{2}, \qquad h_{t+1} = \begin{cases} A_t, & U_t = 1, \\ h_t, & U_t = 0. \end{cases}$$

$$m_t = (1 - \eta) \mathcal{B}(s_t, \rho_t) + \eta h_t, \qquad p_t = (1 - \epsilon_t)\big[(1 - \eta) \mathcal{B}(s_t, \rho_t) + \eta h_t\big] + \epsilon_t/2.$$

In models 12–13, h_t never updates ξ_t; ξ_t never updates h_t. The latent state always learns from valid category feedback, including when the bias state is action-driven.

Likelihood, masks, causal order and exact limits

For history-only models 1–4, $\mathcal{V}_d = \{t : U_t K_t = 1\}$ (duration is unused). Sensory models require $D_t U_t = 1$ and a binary current category for lapse. Models 8, 9 and 12 explicitly also mask $K_t = 0$; the other sensory evaluators expect a valid category on scored rows and reject invalid lapse input. On valid task data:

$$J_d = -\sum_{t \in \mathcal{V}_d} \big[y_t \log \bar{p}_t + (1 - y_t) \log(1 - \bar{p}_t)\big], \quad n_d = |\mathcal{V}_d|,$$

$$\mathcal{L}_{\text{pooled}} = \frac{\sum_d J_d}{\sum_d n_d}, \qquad \hat{\Theta} = \arg\min_{\Theta} \mathcal{L}_{\text{pooled}}; \qquad \mathcal{L}_{\text{equal}} = \frac{1}{D} \sum_d \frac{J_d}{n_d} \quad \text{(sensitivity only)}.$$

D here is the number of sessions, distinct from trial-validity D_t. Zero-valid sessions contribute zero to pooled count/loss; the equal-session objective requires every $n_d > 0$. The total pooled count must be positive. Only training choices enter optimization. All models predict, score, then update; all states reset at each session, including validation/test sessions. Parameters do not reset inside a session. The only current-category dependence before feedback is the common lapse selection.

Exact explicitly constructed boundaries: $\alpha = 1$ reduces model 2 to 1 and 4 to 3; $\eta = 0$ reduces 8/9 to 5 and 12/13 to 11; $b_h = 0$ reduces model 11's block marginal to 10; $\delta_0 = 0$ gives $\rho_t = 1/2$ in M2. Free logistic/softplus coordinates approach these boundaries but do not attain them. No prototype learning occurs in models 10–13. Scope: active 13-model registry in post_analysis/expanded_model_comparison/config.py (2 November 2026). Pending dynamic action-kernel bias and M3/M4 are outside this set. Sources checked: docs/models; shared sensory, WSLS/action and previous-bias evaluators; M2 session and capped-tail HSMM implementations. Equations specify the scientific forward models and fitting likelihood, not optimizer algorithms.